Power Series and Taylor Series

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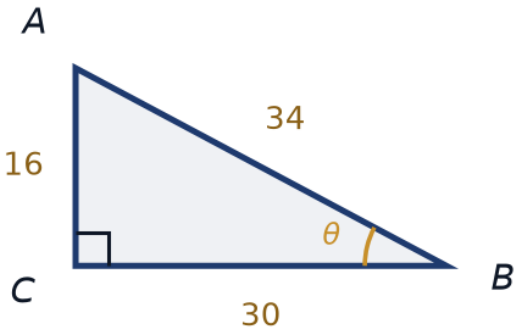
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Taylor and Maclaurin series

Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.

2 [Foundational] • Target: Taylor and Maclaurin series

Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.



3 [Foundational] • Target: Taylor and Maclaurin series

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.

4 [Foundational] • Target: Interval of convergence (including endpoints)

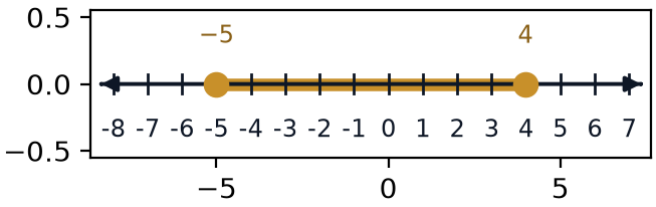
Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.

5 [Foundational] • Target: Using known series

Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.

6 [Foundational] • Target: Using known series

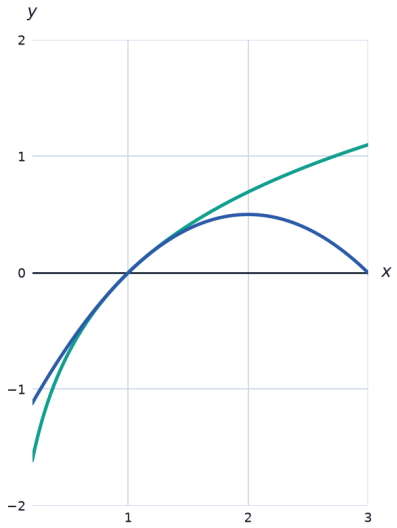
Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).



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7 [Foundational] • Target: Taylor series centered at a nonzero point

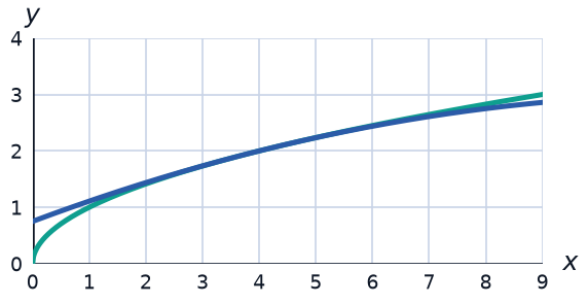
Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Taylor series centered at a nonzero point

Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.



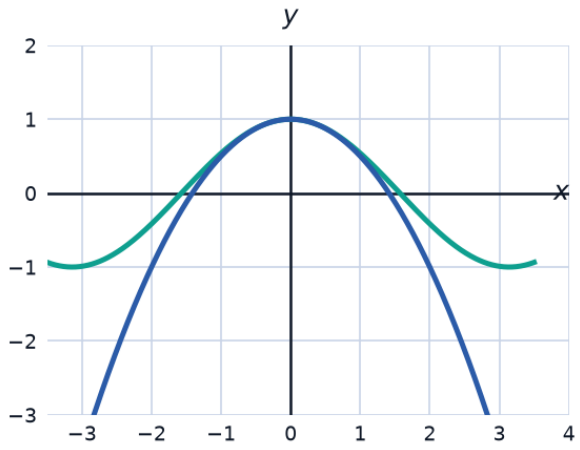
9 [Intermediate] • Target: Error estimation with Taylor polynomials

Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

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10 [Intermediate] • Target: Visualizing a Taylor approximation

The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.



11 [Intermediate] • Target: Differentiating and integrating power series

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

12 [Intermediate] • Target: Differentiating and integrating power series

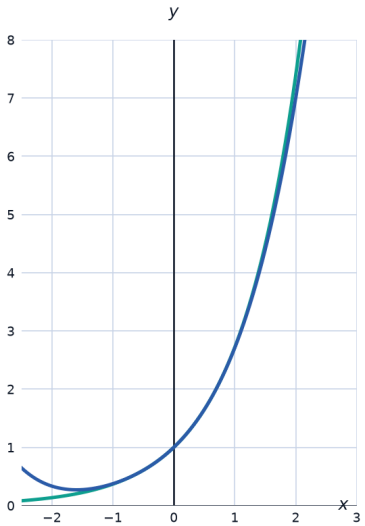
Using the Maclaurin series $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x \, dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

13 [Intermediate] • Target: Using Taylor series to evaluate a limit

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

14 [Intermediate] • Target: Using Taylor series to evaluate a limit

Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.



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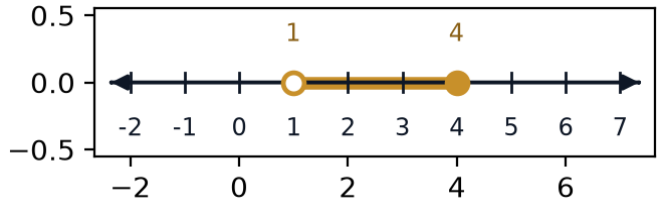
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 7.02 Integration by Parts

An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.

16 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 8.01 Area Between Curves

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.



17 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 8.02 Volumes of Revolution

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.

19 [Challenging] • Target: Radius and interval of convergence • Cross-Lesson: 9.01 Separable Differential Equations

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.

20 [Challenging] • Target: Taylor and Maclaurin series • Cross-Lesson: 1.01 Precalculus Review for Calculus

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0, 1]$ and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Radius and interval of convergence

Interval of convergence of $\sum \frac{(-3)^n (x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R = 1/3$, interval $(5/3, 7/3]$

b. $R = 1/3$, interval $[5/3, 7/3)$

c. $R = 1/3$, interval $[5/3, 7/3]$

d. $R = 3$, interval $(-1, 5]$

22 [Foundational] • Target: Taylor and Maclaurin series

The interval of convergence always contains at least the point:

a. $x = R$

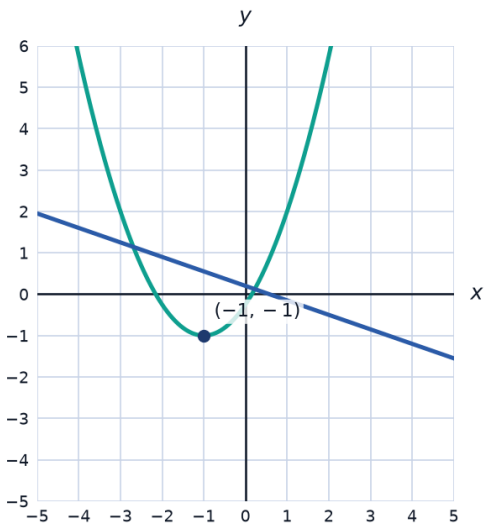
b. $x = 1$

c. $x = -1$

d. $x = 0$

23 [Foundational] • Target: Interval of convergence (including endpoints)

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$



a. 1

b. $f(x)$

c. $n!$

d. 0

24 [Foundational] • Target: Using known series

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21

c. 1/19

d. 0.01

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25 [Foundational] • Target: Taylor series centered at a nonzero point

If $|f^{(4+1)}(c)| \le 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. 0.00492075

c. 0.00246038

d. 0.0273375

26 [Foundational] • Target: Error estimation with Taylor polynomials

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:

a. $1 + x + \frac{x^2}{2}$

b. e

c. $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

d. $1 + x + \frac{x^2}{2} - \frac{x^3}{3}$

27 [Foundational] • Target: Visualizing a Taylor approximation

Taylor series of $1/(1 - x)$ at $x = 0$:

a. $\sum x^n/n!$

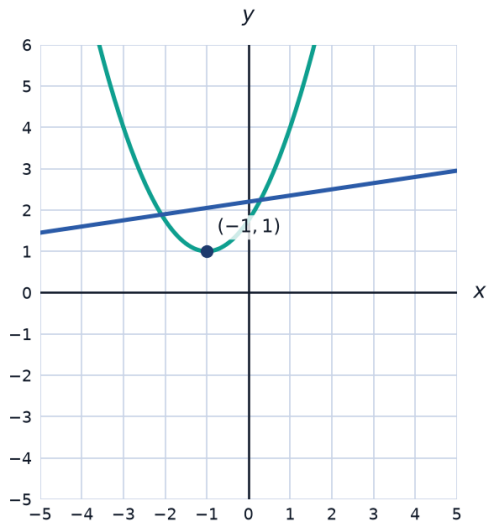
b. $\sum (-1)^n x^n$

c. $\sum x^{n+1}$

d. $\sum x^n$ for $|x| < 1$

28 [Foundational] • Target: Differentiating and integrating power series

Use alternating series estimation: $|\cos 1 - 1 + 1/2| \le |-1/4!\,| =$



a. 1/12

b. 1/120

c. 1/6

d. 1/24

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Using Taylor series to evaluate a limit**

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x = 0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$

c. $0.1^2/2$

d. $1/24$

30 **[Intermediate]** • **Target: Radius and interval of convergence**

Radius of convergence of $\sum x^n/n$ is $R =$

a. 0

b. e

c. ∞

d. 1

31 **[Intermediate]** • **Target: Taylor and Maclaurin series**

A power series $\sum c_n x^n$ with radius of convergence R converges absolutely for:

a. no x

b. all x

c. |x|<R

d. $|x| > R$

32 **[Intermediate]** • **Target: Interval of convergence (including endpoints)**

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$

a. 2

b. 1

c. $-1/2$

d. $1/2$

33 **[Intermediate]** • **Target: Using known series**

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum n x^{n-1}/(n+1)$

c. $1/(1-x)$

d. $\ln(1-x)$

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34 [Intermediate] • Target: Taylor series centered at a nonzero point

Approximate $\int_0^1 \frac{\sin x}{x} dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

35 [Intermediate] • Target: Error estimation with Taylor polynomials

Find radius of convergence: $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$:

a. $1/4$

b. $1/2$

c. ∞

d. 4

36 [Intermediate] • Target: Visualizing a Taylor approximation

If the radius of convergence of $\sum a_n x^n$ is R, then the radius of $\sum a_n x^{2n}$:

a. R

b. R^2

c. $\sqrt{R}$

d. $2R$

37 [Intermediate] • Target: Differentiating and integrating power series

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Sum $\sum_{n=1}^{\infty} n(n-1)x^n/(n!)$ simplifies using e^x series to:

a. $e^x(x^2 - x)$

b. xe^x

c. $x^2 e^x$

d. $x^2 e^x/2$

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39 [Challenging] • Target: Radius and interval of convergence • Cross-Lesson: 9.01 Separable Differential Equations

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

b. are zero

c. exist

d. diverge

40 [Challenging] • Target: Taylor and Maclaurin series • Cross-Lesson: 1.01 Precalculus Review for Calculus

Power series $\sum \frac{(x-2)^n}{3^n}$: centre and radius:

a. centre 0, $R = 3$

b. centre 3, $R = 2$

c. centre 2, $R = 1/3$

d. centre 2, $R = 3$

41 [Challenging] • Target: Interval of convergence (including endpoints) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Using power series, compute $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$:

a. 1/2

b. 1

c. 0

d. 2

42 [Challenging] • Target: Using known series • Cross-Lesson: 11.01 Sequences and Series

Power series $\sum_{n=0}^{\infty} n(x-1)^n$. Radius of convergence:

a. 0

b. 1

c. 2

d. ∞

43 [Challenging] • Target: Taylor series centered at a nonzero point • Cross-Lesson: 12.01 Convergence Tests

Power series centred at a: the interval of convergence always includes:

a. $(-\infty, \infty)$

b. $x = a$

c. the endpoint a

d. $x = 0$

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44 [Challenging] • Target: Error estimation with Taylor polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Hypergeometric series ${}_2F_1(a,b;c;x)=\sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n n!}x^n$ (Pochhammer symbol). Radius of convergence:

a. depends on a, b, c

b. always ∞

c. always 1

d. always 0

45 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

If a function equals its Taylor series everywhere, it is called:

a. analytic

b. polynomial

c. differentiable

d. continuous

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